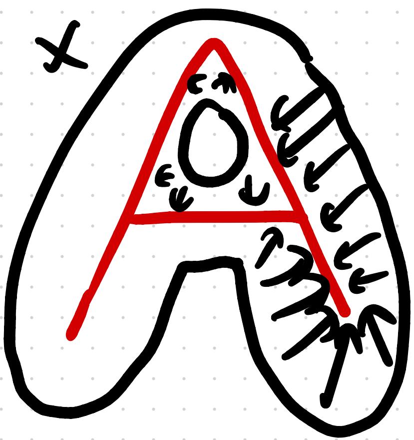
Let X, Y be topological spaces. Two (continuous) maps $f: X \to Y$ and $f': X \to Y$ are **homotopic** if one map can be continuously deformed to the other.

Formally, $f \simeq g$ if there exists a continuous function $F: X \times [0,1] \to Y$ s.t. $F(x,0) = f(x)$, $F(x,1) = f'(x)$ $\forall x \in X$.

Two spaces X, Y are **homotopy equivalent** if there exist maps $f: X \to Y$, $g: Y \to X$ s.t. $f \circ g$ and $g \circ f$ are both homotopic to the identity map.

Let X be a topological space and A a subspace. A **deformation retract** of X onto A is a map

$$F : X \times [0,1] \to X \quad \text{s.t.} \quad F(x,0) = x, \; F(x,1) \in A \quad \forall x \in X$$
$$F(a,t) = a \quad \forall a \in A$$



X deformation retracts onto A.

<u>Prop</u>: If X deformation retracts onto A, then X and A are homotopy equivalent.

Theorem: If Δ and Δ' are simplicial complexes such that $|\Delta|$ and $|\Delta'|$ are homotopy equivalent, then $H_\bullet(\Delta) \cong H_\bullet(\Delta')$.

Simplicial manifolds

Recall that a **manifold** of dimension d is a topological space where every point has a neighborhood homeomorphic to $\mathbb{R}^d$. A **simplicial manifold** is a simplicial complex homeomorphic to a manifold.

Manifolds: S^{n-1} spheres

$\mathbb{RP}^n$ real projective space

$\mathbb{CP}^n$ complex projective space

$S^1 \times S^1$ torus

Given a simplicial complex Δ and a face $\sigma \in \Delta$, define the **link** of σ to be

$$\operatorname{link}_\Delta \sigma = \{\tau \in \Delta : \tau \cup \sigma \in \Delta,\ \tau \cap \sigma = \emptyset\}$$

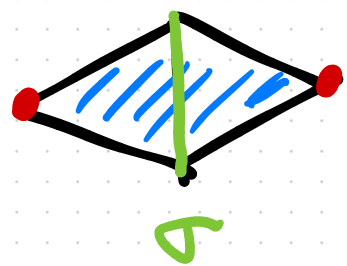
We define the **star** of σ to be

$$\operatorname{star}_\Delta \sigma = \{\tau \in \Delta : \tau \cup \sigma \in \Delta\}$$

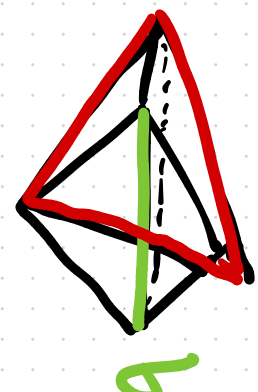
star

link

$\operatorname{link}_\Delta \sigma \approx S^1$



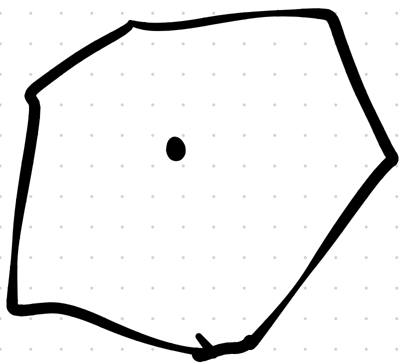
$\operatorname{link}_\Delta \sigma \approx S^0$



$\operatorname{link}_\Delta \sigma \approx S^1$

Proposition: If Δ is a simplicial d-manifold, then for all nonempty $\sigma \in \Delta$,

$$H_\bullet(\mathrm{lk}_\Delta \sigma) \cong H_\bullet(S^{d-1-\dim \sigma})$$

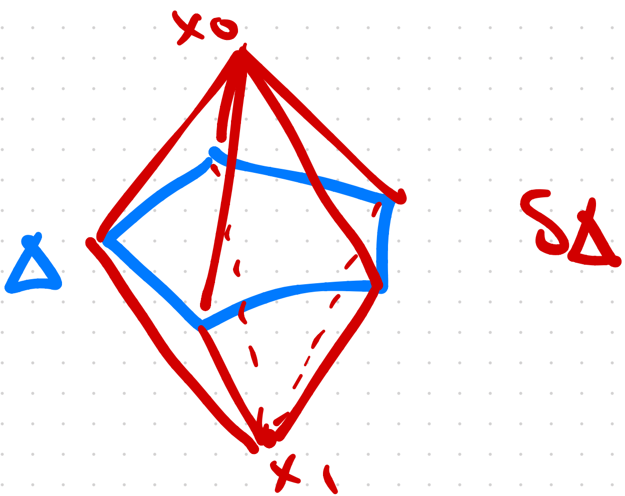


A space with the same homology as a sphere is called a homology sphere.

Not always true that $\mathrm{lk}_\Delta \sigma \approx S^{d-1-\dim \sigma}$ (when $\dim \geq 5$)

Given a simplicial complex Δ, the suspension of Δ is

$$S\Delta = \Delta \cup \{x_0 \cup \sigma : \sigma \in \Delta\} \cup \{x_1 \cup \sigma : \sigma \in \Delta\}$$



<u>Theorem</u>: If Δ is a homology sphere, then $S^2\Delta$ is homeomorphic to a sphere.

Let Δ be a homology 3-sphere which is not a sphere.

Then $S\Delta$ is a homology 4-sphere, not a sphere

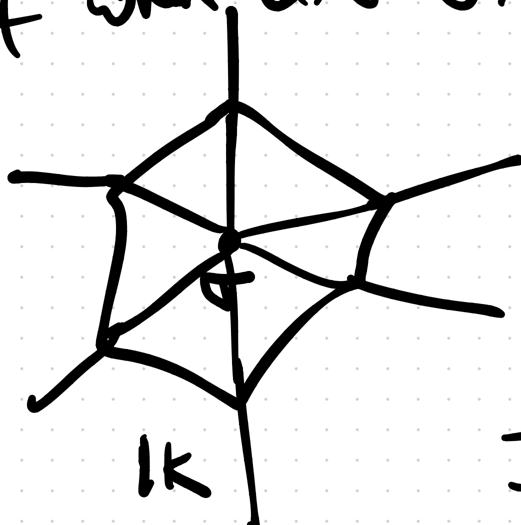
Then $S^2\Delta$ is a 5-sphere.

In $S^2\Delta$, the link of x_0 is $S\Delta$.

Proposition: If Δ is a simplicial d-manifold, then for all nonempty $\sigma \in \Delta$

$$H_\bullet(\operatorname{lk}_\Delta \sigma) \cong H_\bullet(S^{d-1-\dim\sigma})$$

Proof when $\dim\sigma = 0$.



$$(\operatorname{lk}_\Delta \sigma) \times [0,\infty) \big/ \operatorname{lk}_\Delta \sigma \times 0 \approx \mathbb{R}^d$$

If we puncture the origin,

$$(\mathrm{lk}_\sigma \sigma) \times (0, \infty) \approx \mathbb{R}^d / 0$$

$$(\mathrm{lk}_\sigma \sigma) \times \mathbb{R} \approx S^{d-1} \times \mathbb{R}$$

For any space X, $X \times \mathbb{R}$ is homotopy equivalent to X.

Thus $\mathrm{lk}_\sigma \sigma$ and S^{d-1} are homotopy equivalent.

<u>Corollary</u>: For Δ a simplicial d-manifold and all nonempty $\sigma \in \Delta$.

$$\sum_{i \geq -1} (-1)^i f_i(\mathrm{lk}_\Delta \sigma) = (-1)^{d-1-\dim\sigma}$$

reduced Euler characteristic of $S^{d-1-\dim\sigma}$

$$\sum_{\tau \in \mathrm{lk}_\Delta \sigma} (-1)^{\dim\tau} = (-1)^{d-1-\dim\sigma}$$

$$\sum_{\Delta \ni \sigma' \supset \sigma} (-1)^{\dim\sigma' - \dim\sigma - 1} = (-1)^{d-1-\dim\sigma}$$

$$\sum_{\Delta \ni \sigma' \supset \sigma} (-1)^{\dim\sigma'} = (-1)^d$$